

VIVEKANAND S NAIK

Software Engineer | Full Stack & Backend Developer | AI/ML Systems

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SUMMARY

Computer Science undergraduate skilled in building full-stack and AI-powered backend systems with Python, Java, and modern web frameworks. Experienced designing REST APIs, integrating computer vision/ML models into production-style pipelines, and architecting scalable server-side systems. Strong foundation in DSA, databases, and system design.

TECHNICAL SKILLS

Languages: Java (Core & Advanced), Python, JavaScript, SQL, C

Backend & Web: Spring Boot, Flask, FastAPI, REST API Design, HTML5, CSS3

AI / Machine Learning & Computer Vision: TensorFlow, Keras, PyTorch, OpenCV, MediaPipe, dlib, face_recognition, YOLO, LangChain

Databases: MySQL, PostgreSQL, MongoDB

Tools & Platforms: Git, GitHub, IntelliJ IDEA, Eclipse, VS Code, Firebase, Postman

Core Concepts: Data Structures & Algorithms, OOP, System Design Fundamentals, REST APIs, Authentication

PROJECTS

Intelligent AI Surveillance & Real-Time Vision Analytics Platform

AI/ML • COMPUTER VISION • BACKEND

Multi-model real-time video intelligence system unifying identity verification, behavior analysis, and security alerting.

- Built a real-time surveillance platform unifying facial recognition, activity analysis, gaze tracking, and spoof detection into a single multi-model inference pipeline.
- Engineered anti-proxy/liveness detection and I3D-based suspicious-activity detection on live video streams.
- Exposed functionality via REST APIs (Flask/FastAPI) and built a Pandas/Plotly analytics dashboard for real-time monitoring.

Tech Stack: Python, OpenCV, TensorFlow/Keras, PyTorch, MediaPipe, face_recognition, dlib, Flask, FastAPI, Pandas, Plotly

CivicEye | AI-Driven Smart City Infrastructure Monitoring Platform

COMPUTER VISION • GEOSPATIAL • BACKEND

Automates detection and geo-mapping of civic infrastructure issues to replace manual inspection with AI-driven reporting.

- Built an AI-driven platform that automatically detects civic issues (potholes, garbage, damaged roads, waterlogging) from image/video data using YOLO object detection.
- Integrated computer vision with geolocation/GIS mapping and built REST API backend services with a reward-and-penalty rule engine.
- Designed reporting/analytics features to help authorities prioritize issues by severity and location.

Tech Stack: Python, OpenCV, TensorFlow/PyTorch, YOLO, FastAPI, Flask, MongoDB/MySQL, React, GIS/Mapping APIs

ParallaxAI | LLM-Powered Intelligent Assistant & Workflow Automation Platform

GENERATIVE AI • LLM • BACKEND

Context-aware AI assistant with a modular, backend-first architecture supporting RAG and extensible AI workflows.

- Designed an LLM-powered assistant with a modular, backend-first architecture (FastAPI/Flask) integrating external APIs and AI orchestration via LangChain.
- Implemented Retrieval-Augmented Generation (RAG) with a vector database, plus authentication and database integration for scalable, multi-user AI workflows.

Tech Stack: Python, LangChain, LLM APIs, FastAPI, Flask, Vector Database (RAG), React, REST APIs

EDUCATION

Bachelor of Engineering, Computer Science

2023 – 2027

Srinivas University Institute of Engineering and Technology, Mangalore, Karnataka | CGPA: 9.1

Pre-University Course (PCMC)

2023

Kendriya Vidyalaya, Karwar, Karnataka | 70%

CERTIFICATIONS

- IBM Certified — SQL & RDBMS
- IBM Certified — Java Programming
- Ethnotech Certified — Data Science

ACHIEVEMENTS

- Won a competitive off-campus hackathon by designing and building an innovative solution under tight time constraints.
- Organized a college-level hackathon (registrations, evaluation, logistics, judging) and anchored technical/cultural events, showcasing leadership and public speaking.

ADDITIONAL INFORMATION

Areas of Interest: AI/ML, Full Stack Development, Data Science **Interests:** Travelling, Teaching, Cooking